

“IOT BASED COALMINE SAFETY MONITORING WITH HAZARD DETECTION & REAL TIME ALERT”

A project report Submitted to

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY,
HYDERABAD**

In partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the PROJECT REPORT entitled “IOT BASED COALMINE SAFETY MONITORING WITH HAZARD DETECTION & REAL TIME ALERT”, is being submitted by **B.NANDINI (23TR1A0407), G.RUCHITHA (24TR5A0402), S.PRANENDRA (23TR1A0452), R.GOURIPRIYA (23TR1A0447), G.AKSHAY KUMAR(23TR1A0420),K.BHARATH KUMAR(23TR1A0431),P.KEERTHANA(23TR1A0443),V.SUDHEE R(23TR1A0461)** in partial fulfillment of the requirements for the award of Bachelor of Technology in Electronics and Communication Engineering at Sree Chaitanya Institute of Technological Sciences, Karimnagar is a Bonafide work carried out by them.

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DECLARATION

We hereby declare that the work which is being presented in this dissertation entitled, “**IOT BASED COALMINE SAFETY MONITORING WITH HAZARD DETECTION & REAL TIME ALERT**”, submitted towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**, SCITS, Karimnagar, is an authentic record of our own work carried out under the guidance of **Mrs.K.NEETHIKA, Assistant professor**, Department of ECE, SCITS, Karimnagar.

To the best of our knowledge and belief, this project bears no resemblance with any report submitted to SCITS or any other University for the award of any degree or diploma.

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ABSTRACT

Coal mines are one of the most important industries in the country, as they are used as fuel in the steel and cement industries to extract iron from the stone and create cement. The coal mining industry is known for its hazardous working environment, requiring stringent safety measures to protect miners and prevent accidents. The coal mine safety and monitoring project provides a comprehensive solution to enhance safety within coalmines. The objective of this project is to continuously monitor critical parameters such as temperature, gas concentration to ensure a safe working environment in coal mines. The continuous monitoring of temperature, gas concentration, along with remote communication capabilities and alerting mechanisms, contribute to minimizing the risk of accidents, improving response times, and overall safety standards within coal mining operations.

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CHAPTER 1

INTRODUCTION

Industrial safety, particularly in the coal mining industry, is of paramount importance due to the hazardous conditions miners face daily. Underground mining, while vital to energy production, comes with inherent risks such as suffocation, gas poisoning, and dangerous explosions due to the presence of flammable gases like methane (CH₄) and carbon monoxide (CO). These risks make it critical to monitor air quality, temperature, and humidity in real time to ensure the safety of miners. Additionally, a significant cause of mining accidents is miners losing consciousness due to falls or other injuries, often without immediate access to medical attention. Furthermore, not all workers adhere to safety protocols, such as wearing helmets, which further compounds the risks they face. The need for a comprehensive, real-time monitoring system is therefore crucial to mitigate these dangers. The proposed system uses IoT technology to provide continuous monitoring of environmental conditions in the mine and ensure rapid response in case of emergencies, such as gas leaks, high temperature or a worker's fall.

1.1 IOT

The Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human to-computer interaction. IoT has evolved from the convergence of wireless technologies, micro-electro mechanical systems (MEMS), micro services and the internet. The convergence has helped tear down the silo walls between operational technology (OT) and information technology (IT), allowing unstructured machine-generated data to be analyzed for insights that will drive improvements.



Fig 1.1 IOT

Wireless Communication Protocols

- **Wi-Fi:** Used for high-speed data transmission in smart homes and offices.
- **Bluetooth & BLE (Bluetooth Low Energy):** Ideal for short-range device communication, such as wearable gadgets.
- **Zigbee & Z-Wave:** Low-power protocols often used in smart home automation.
- **NFC (Near Field Communication):** Enables close-proximity interactions, like contactless payments.

Cellular Networks (2G/3G/4G/5G)

- IoT devices use cellular connections to transmit data over large distances— think smart meters and vehicle tracking systems.

LPWAN (Low Power Wide Area Networks)

- **LoRaWAN & NB-IoT:** Designed for devices that need minimal energy consumption but long-range communication, such as environmental sensors.

Cloud & Edge Computing

- Devices send data to cloud platforms for storage and analysis.
- Some IoT devices process data locally using edge computing, reducing latency for real-time applications.

Machine-to-Machine (M2M) Communication

- Devices interact directly without human intervention, like industrial automation systems.

1.2 THINGSPEAK

In order to connect an object to the IoT, several things are needed in the hardware and software realm. First of all, if one wishes to go beyond simply connecting data from a computer, objects to gather (sensors) or receive (actuators) data are necessary. For example, a digital thermometer can be used to measure temperature. In this case, the data needs to be uploaded to a network of connected servers which run applications . Such a network is commonly referred to as ‘the cloud’. The cloud utilizes the process of virtualization, meaning that several physical servers can be connected and used in tandem, but appear to the user as one machine (despite that at the physical level, the machines function independently).

This method of computing thus allows changes to be made to the ‘virtual’ server (such as software updates or changes in storage space) much easier than before.

In this case, an object will connect to the cloud through a (possibly wireless) Internet connection to upload or receive data. Objects to be connected are typically augmented with either sensors or actuators.

A sensor is something that tells us about our environment. Think of a temperature sensor, or even the GPS receiver on your mobile phone. Actuators are something that you want to control. Things like thermostats, lights, pumps, and outlets.

The IoT brings everything together and allows us to interact with our things and, even more interestingly, allows things to interact with other things. For the purpose of connecting an object to the IoT, we focus on the Thing Speak API.

The Internet of Things (IoT) is all about connecting everyday objects to the internet, enabling them to collect, exchange, and act on data. Imagine smart home devices that can adjust the temperature or lights automatically, or wearable fitness trackers that monitor your health in real time.

IoT is transforming industries like healthcare, manufacturing, and transportation by making systems more efficient, predictive, and intelligent. For instance, smart cities use IoT to manage traffic, energy, and security seamlessly.

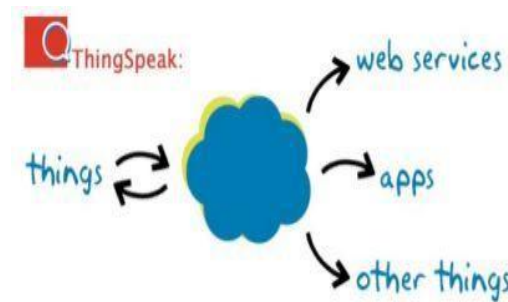


Fig 1.2 Things representing itself as a cloud interface

The interface provides simple communication capabilities to objects within the IoT environment, as well as interesting additional applications (such as Thing Tweet, which will be further discussed in a later section). Moreover, Thing Speak allows you to build applications around data collected by sensors. It offers near real-time data collection, data processing, and also simple visualizations for its users.

Data is stored in so-called channels, which provides the user with a list of features each channel allows you to store up to 8 fields of data, using up to 255 alphanumeric characters each. There are also 4 dedicated fields for positional data, consisting of:

Description, Latitude, Longitude, and Elevation. All incoming data is time and date stamped and receives a sequential ID. Once a channel has been created, data can be published by accessing the Thing Speak API with a ‘write key’, a randomly created unique alphanumeric string used for authentication.

Channel data in case it is set to keep its data private (the default setting). Channels can also be made public in which case no read key is required. According to the Thing Speak website, the API works as noted in Figure 1. Essentially, ‘things’ are objects that

are given sensors to collect data.

Data is sent and received via simple “Hypertext Transfer Protocol” (HTTP) POSTs, much like going to a web page and filling out a form. This communication happens through plaintext, JSON or XML. The data is then uploaded to the cloud and from there can be used for a variety of purposes. In turn, data (such as commands or choosing certain options) can be gathered and communicated to the cloud, which in turn sends these messages to the object. A deeper level of what occurs, especially on the server side. When a device sends data through a HTTP request (communication), it is processed by the IoT service (in this case Thing Speak), which communicates with a virtual server.

Both the server and the IoT service communicate directly with the application. Finally, at all levels of communication from the device to the application there is both requirements regarding security and management of the data transfer. Unfortunately, Thing Speak does not document how the specific parts of the diagram are handled on a technical level. Given enough time and expertise, it should however be possible to answer this by looking into the (open source) code base.

For this report, we take the example of connecting a doorbell (herein referred to as the ‘tweeting doorbell’) to the Thing Speak web service which will send a message to Twitter whenever the doorbell was rung.

The connection from Thing Speak to Twitter is handled by the application 'Thing Tweet', a script independent from the Thing Speak API but available when using the API from the Thing Speak web service.

Messages from Thing Speak are communicated to Twitter via an HTTP POST. This feature actually enables users to bypass displaying data in a channel and directly upload it to Twitter, if desired.

Thing Speak is an IoT analytics platform that allows users to collect, visualize, and analyze live data streams in the cloud. It enables IoT devices to send data, create instant visualizations, and even perform advanced analytics using MATLAB.

Key Features of Thing Speak

- **Data Collection:** Devices can send sensor data privately to the cloud.
- **Visualization:** Users can create real-time graphs and dashboards.
- **Analysis:** MATLAB integration allows for deeper insights and predictive analytics.
- **Automation:** IoT analytics can be scheduled or triggered by events.
- **Integration:** Works with platforms like Arduino, Raspberry Pi, ESP8266, and LoRaWAN

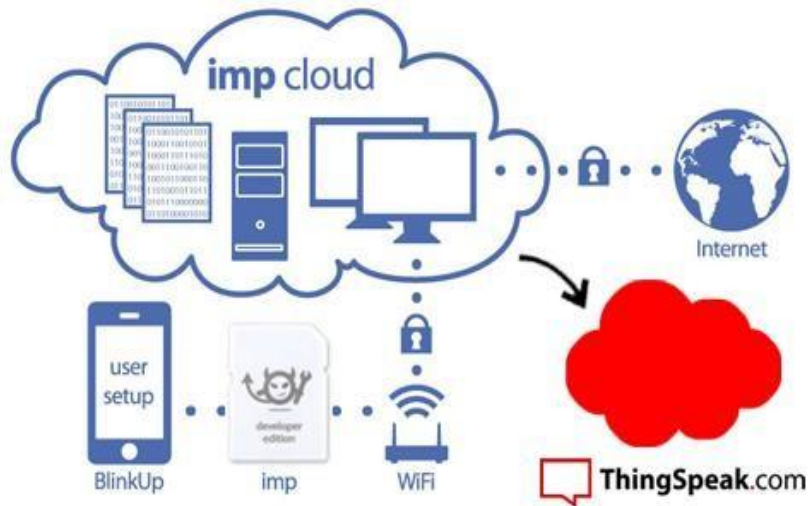


Fig 1.3 Thing Speak

Advanced Features of ThingSpeak

- **Real-Time Data Collection:** Supports HTTP and MQTT protocols for seamless data transmission.
- **Data Visualization:** Built-in tools for creating real-time charts and graphs.
- **MATLAB Integration:** Enables complex data processing and machine learning.
- **Event Scheduling & Alerts:** Automate actions based on predefined triggers.
- **RESTful & MQTT APIs:** Allows easy connectivity with IoT devices.
- **Public & Private Channels:** Users can choose to share data publicly or keep it private.

Applications of Thing Speak

- **Environmental Monitoring:** Used for tracking air quality, weather conditions, and pollution levels.
- **Energy Management:** Helps monitor energy usage and predict power consumption.
- **Smart Farming:** Enables remote monitoring of soil conditions and crop health.
- **Traffic Analysis:** Used for real-time traffic monitoring and congestion prediction.

All incoming data is time and date stamped and receives a sequential ID. Once a channel has been created, data can be published by accessing the Thing Speak API with a 'write key', a randomly created unique alphanumeric string used for authentication. Consequently, a 'read key' is used to access channel data in case it is set to keep its data private (the default setting). Channels can also be made public in which case no read key is required. According to the Thing Speak website, the API works. Essentially, 'things' are objects that are given sensors to collect data.

CHAPTER-2

LITERATURE SURVEY

Wireless Sensor Networks in Mines: Several studies have explored the use of wireless sensor networks (WSNs) for real-time monitoring in mines. For example, "Real-time Monitoring in Underground Mines Using Wireless Sensor Networks"(2019) highlights the effectiveness of WSNs in providing continuous monitoring of environmental factors like gas concentrations and temperature. These systems can offer better coverage and are easier to implement than traditional wired systems, making them more suitable for hazardous environments.

IoT in Mining Safety: Significant trend in recent research is the integration of IoT technology for monitoring and alerting purposes. "IoT-based Mining Safety System" (2021) discusses the potential of IoT for monitoring gases, temperature, and other environmental parameters in Realtime, and emphasizes the need for automation in emergency response. These systems enable miners to send alerts in the event of gas leaks or health problems, which could help save lives in emergency situations.

Fall Detection Systems: Several studies have focused on fall detection mechanisms using wearable sensors. Research in "Wearable IoT Devices for Worker Safety"(2020) outlines how accelerometers and gyroscopic sensors can accurately detect falls in hazardous work environments like mining. These studies underline the importance of immediate alert systems for improving response times and preventing fatalities.

Helmet Monitoring: A study on "Helmet Monitoring Systems for Worker Safety" (2018) explores how wearable devices can be used to ensure that miners are wearing safety equipment like helmets. The study suggests that integrating helmet sensors with a central monitoring system can alert supervisors when workers are not complying with safety protocols.

Challenges in Implementation: Despite the promising advancements, challenges remain in implementing such systems effectively. For instance, research by "Challenges in Deploying IoT Systems in Underground Mines" (2022) discusses issues like signal interference, battery life, and environmental hazards that can affect the

performance of IoT-based systems. Addressing these challenges is crucial to ensure the reliability of safety systems in such a harsh and dangerous environment. It is characterized by its low power consumption, cost-effectiveness, and ability to offer intelligent monitoring and safety measures.

The system comprises a wireless network comprising several nodes. This network may be conveniently installed in underground mines and offers an efficient monitoring and safety system for underground coal miners. Specifically, it facilitates instantaneous data transmission between miners and the surface control center using extremely secure and dependable wireless sensor nodes. The placement of this system in the mine creates an issue when miners are out of range of the system. This system just monitors the environmental conditions inside the underground mine, but it does not monitor the health conditions of the miners. Yongping Wu and Guo Feng use the Bluetooth wireless transmission method to execute coal mine monitoring. Bluetooth technology serves as a universal standard for short-range wireless communication worldwide. Its purpose is to provide a shared, low-power, low-cost wireless interface and software system for controlling devices.

This article provides an overview of the development history, technological characteristics, and structure of the protocol stack of Bluetooth technology. It also proposes solutions for the difficulty of developing the Bluetooth host controller interface (HCI) wireless communication. Zhenzhen Sun introduced the DCS Coal Mine Monitoring System. The RS485 bus topology is designed to provide multipoint and bidirectional communication. This monitoring system maybe created utilizing widely It offers the benefits of a simple circuit design and affordable expenses. However, the presence of a master-slave structure in the network makes it challenging to ensure the durability of the network's structure. In addition, the data transfer distance is restricted and the real time performance is subpar. Jing jiang Song and Yingli Zhu suggested an automated network. This system is designed to monitor the safety of coal mines and is created using the MSP430F and nRF2401.

The system's sensor groups closely monitor temperature, humidity, and other conditions. The microcontroller transmits the measured parameters to the wireless communication module. The gathered data is sent via cable to a remote monitoring center. The

issue with this approach is that the hardware is located within the coal mines. When a natural disaster or a roof collapse occurs, the system is damaged. The typical communication system has low dependability and a short lifespan. The challenging conditions inside the mine make it very difficult to install and maintain the system. Another issue is the high level of noise in coal mines, which might hinder effective communication between miners and the system, especially if they are far apart.

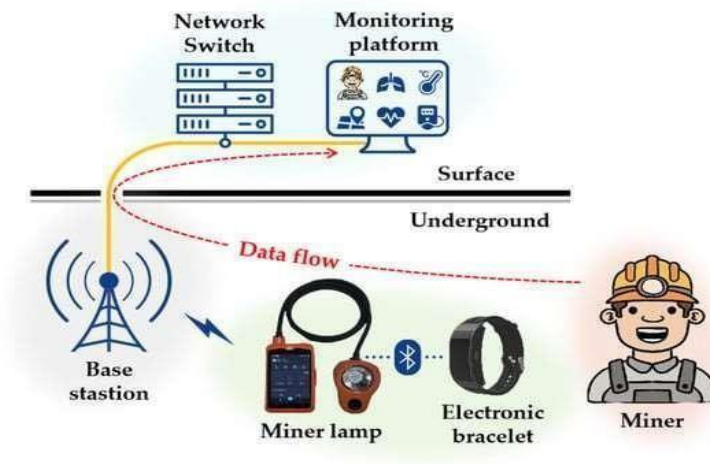


Fig No 2.1 IOT based coal mine safety monitoring system architecture

PROBLEM IDENTIFICATION

Surveillance cameras were always used for tracking.

- It requires humans to monitor and spot potentially dangerous circumstances.
- The current method produces instabilities.
- They could not control the death of mine workers

OBJECTIVES

The identification of various toxic substances in the industrial area.

- Establishing communication between devices and Wi-Fi.
- Creation of a wireless sensor network.
- Creation of a mechanism for real-time observation.
- Save the health condition of mine workers.

METHODOLOGY

Our undertaking is divided into two parts. The first one involves keeping track of the health of mine employees, and the second one involves detecting poisonous gas in coal mines. We are utilizing a gas monitor, MEMS sensor, and pulse sensor. The NODE MCU will enable and notify the person through the buzzer and information shown in LCD if any gas sensor value exceeds the cutoff range at the spot. Members of the Monitoring area can view and be aware of the state of the Mine workers as well as refresh the information that has been gathered using the Zig bee module and IOT.

APPLICATIONS

- It is used to prolong the life of coalminers and it is constantly observable.
- If any urgent scenario arose, a note was immediately sent via mail or message.
- Establishing communication between sensors and Wi-Fi.
- Detecting various toxic chemicals in the industrial area.

V. Chandra sekhar...,(2018) presented a real-time flood tracking and alert

management system built on the internet of things. The system is extremely beneficial in terms of safeguarding humans and animals' lives. The suggested model is widely used for tracking water levels and flow variations in rivers, and it can also be used to measure water levels at dams and reservoirs. The measured values are changed on the web server on a regular basis, which is very useful for sending flood warnings to consistent authority and 14 people for quicker action. The initiative consists primarily of wireless sensor units known as motes, which are put along river beds to measure water quality. Each Node is linked to a cellular modem. The processed data is sent via GPRS from the relevant node to the alarm management system. An application programme interface (API) for Google Spreadsheets was developed, and this API is used as a data recorder. The Google Spreadsheet is linked to free monitoring software known as the warning management system. The warning management System can be used to watch, document, and communicate with people prior to disasters. The monitoring station's data gathering device will benefit from precise and accurate detection of water level. Flood warning information can be posted on LED display

screens for road user and put in strategic areas for safety reasons. Such data should be collected in real time and wirelessly relayed from the measured spot. Solar cells are one potential source of electricity for the sensors and centralized control center. If the Flood Observatory System is fueled by solar cells, it will be simple to deploy and operate. Solar energy will also provide a less expensive source of electricity for the complete system Wan Adlin Harris et al., (2011) and colleagues recommend a text flood alert system This study's findings for the protection of drivers, flood alert information can be displayed on LED displays in key locations. Such information ought to be wirelessly transmitted from the measured location and gathered in real time. One possible source of energy for the sensors and centralized control system is solar cells.

The Flood Observatory System will be straightforward to install and run if it is powered by solar cells. For the entire system, solar energy will also offer a less costly supply of electricity. shows a GSM-based liquid sensor- based remote water level alert device. The system uses the Global System of Mobile Connections (GSM) and Short Message Service (SMS) to send data from sensors to users' mobile phones as it is intended to autonomously monitor water levels. The Easy gate GSM Module, three (3) liquid sensors, a generator, and a Micro Controller Unit (MCU) PIC18F452 makeup the system's components Software for the device consists of (ATTENTION) AT commands in C assembler code. This programme is expected to benefit the neighborhood and act as a safeguard against a water catastrophe in any area that is vulnerable to them.

2.1 EXISTING SYSTEM

In traditional underground mining operations, safety monitoring systems often rely on manual inspections or standalone sensor systems for tracking gases like methane and carbon monoxide. These systems, however, have several limitations:

1. **Manual Monitoring:** Most mines still use manual systems for gas detection and environmental monitoring, where personnel are assigned to check gas level at regular intervals. These checks can be slow, and any delays in detecting hazardous conditions can be catastrophic.
2. **Limited Scope of Safety Features:** Current systems might only detect dangerous gases or environmental factors, but fail to address other critical safety concerns such as monitoring worker health or detecting falls. While some mines use personal protective equipment (PPE) like gas detectors or temperature monitors, there is often no integration between these devices.

3. **No Real-time Alerts:** Most existing systems do not offer real-time alerts to supervisors in case of gas leakage or worker health issues, which could lead to delays in providing help. If a worker falls or loses consciousness, it may take a long time to get help, which increases the risk of severe injury or death.

Several parts makeup this tracking system, including boards (NodeMCU, ADC, and SPI), a Wi-Fi transmitter, various sensors, and other minor electrical parts. Each of these parts is thoroughly discussed in this chapter, along with how it functions. The suggested system fixes the gas sensor modules, temperature sensor, smoke sensor, and switches for the coal mine safety systems. They connect the device to every monitor. To begin, they registered for a Thing Talk account. They primarily have tracking and controlling tools in this system. All of the info from various devices is monitored by the monitoring system. The atmosphere of the coal pit is gas-sensing. The buzzer will sound loudly to alert the mine employees if the gas level rises above the usual level. Continuous uploads of these sensing

ADVANTAGES OF EXISTING SYSTEM

- Using extra instruments, it would be feasible to keep an eye on any potential safety hazards, including gases, dust, vibrations, fire, etc.
- Wi-Fi can also be used to monitor mining activities for things like erosion and water leaks, among other things.
- When wired contact is difficult, this method can be used to transmit other crucial info, making it possible.
- Since the technology allows for simple entry, the control can be managed from the surface itself.

2.2 LIMITATIONS OF EXISTING SYSTEMS

1. Lack of Real-Time Monitoring

Most traditional systems in coal mines are not equipped for real-time tracking. These systems usually operate in a periodic check format, where data is gathered at intervals and reviewed manually.

1.Impact: Delayed response to sudden hazards like gas leaks or cave-ins.

Example: Methane levels may spike between checks, but the danger won't be known until the next reading.

2. Human Dependency and Error : Manual safety checks are still common in many coal mines, especially in developing countries. These include physical gas level measurements and visual inspections.

Limitations:

- Human fatigue, bias, or oversight can lead to missed signs of danger.
- Workers may not have immediate access to protective equipment or alerts if a hazard develops suddenly.

3. Wired and Stationary Sensor Systems

- Many mines use fixed wired sensors, which are limited in their coverage area and lack mobility.

Drawbacks:

- Cannot track mobile hazards (e.g., moving methane pockets).
- Installation and maintenance are difficult in dynamic underground environments.
- Wired systems are vulnerable to damage from blasts or environmental changes (like rock shifts or water seepage).

4. Lack of Integration with Modern Communication Tools

- Traditional safety systems typically lack:
- Wireless connectivity (Wi-Fi, ZigBee, LoRa)
- Cloud integration
- Mobile alert systems (SMS, app notifications)

Consequences:

- Supervisors may not receive alerts when away from the control room.
- Emergency responses may be delayed.
- Data is siloed and not accessible remotely for centralized decision-making.

5. No Worker Tracking or Personal Monitoring

- Existing systems rarely provide real-time tracking of individual miners.
- Difficult to locate workers during emergencies like cave-ins or explosions
- No personal safety monitoring (like wearable gas detectors or heart rate monitors).

2.3 PROPOSED SYSTEM

A coal mine safety system is implemented using IoT platform as a medium to transmit the data. The system is implemented to monitor and control various parameters in the coal mines such as light detection, leakage of gas, temperature and humidity conditions, Fire detection in the coal mine. These all sensors are together considered as one unit and are placed in the coal mines. All the esteems of the sensors are continuously uploaded to the IoT for analysis. Here the gas is continuously monitored if any uncertainties in the level of gas arise, then buzzer is used to alert the workers. In this system LDR sensor is utilized to detect the presence of light. Automatically light gets one and can be controlled using the LED button. In case if any fire occurs in the coal mine, then an alert notification is sent to the mail of the authorized person. Temperature and humidity values are also continuously monitored and displayed on the serial monitor and also in the IoT platform.

Key Features of the Proposed System:

1. Sensors for Hazard Detection:

- MQ-135 Gas Sensor: Detects harmful gases like methane and carbon monoxide.
- DHT11 Sensor: Measures temperature and humidity levels.
- Flame Sensor: Detects the presence of fire or spark.
- Vibration Sensor: Senses shocks or ground movements indicating possible structural collapse.

2. Real-Time Data Monitoring:

- Sensor data is collected continuously and sent to a cloud platform via Wi-Fi or GSM.
- A dashboard or mobile application displays live readings with timestamps.

3. Instant Alert Mechanism:

- If any reading crosses the danger limit, an alert is triggered.
- Alerts are sent via: Buzzer for on-site warning.
- SMS or mobile notification for remote supervisors.

4. Low Power and Cost-Effective:

- The system is designed to run on low power with battery support.
- It uses affordable hardware, making it suitable for small

5. Offline Safety Backup:

- In case of internet or power failure, the system still provides basic safety through the buzzer alarm.
- This ensures that miners are warned even when the network is down, improving overall reliability.

6. Data Logging and Analysis:

- All sensor data is logged in a cloud server or SD card for later review.
- This helps in identifying patterns, maintaining safety records, and meeting regulatory compliance.

7. Scalable and Modular Design:

- The system can be expanded by adding more sensors or nodes as required.
- Suitable for deployment in both small and large mines with different risk levels.

8. Remote Monitoring Capability:

- Supervisors and safety officers can monitor mine conditions from any location using a
- smartphone or web dashboard.
- This enables quick decisions and better coordination during emergencies.

2.4 ADVANTAGES & DISADVANTAGES

ADVANTAGES

1.Real-Time Monitoring:

- Continuously monitors environmental conditions like gas, temperature, fire, and vibrations

2. Instant Alerts:

- Sends immediate alerts via buzzer and mobile notifications to reduce response time during emergencies.

3. Remote Access:

- Enables cloud-based or mobile monitoring, allowing supervisors to track data from anywhere.

4. Multi-Hazard Detection:

- Capable of detecting multiple hazards, improving overall safety coverage in mines.

5. Low Cost & Energy Efficient:

- Uses affordable components and consumes low power, making it suitable for long-term deployment.

6. Data Logging:

- Stores historical data for safety audits, analysis, and reporting.

7. Scalability:

- The system is modular and can be expanded to cover larger mining areas or more sensor types.

8. Easy Maintenance:

- Simple architecture with replaceable modules ensures easy repair and maintenance.

Disadvantages of the Proposed System

1. Internet Dependency (for remote monitoring):

- Cloud and mobile monitoring features require stable internet, which may not be available in all mines.

2. Limited Battery Life:

- If not connected to a constant power source, sensor nodes may require frequent battery replacement or charging.

3. Sensor Limitations:

- Some low-cost sensors may have limited accuracy or sensitivity compared to industrial-grade sensors.

4. Initial Setup Complexity:

- Requires proper calibration and integration of multiple components during installation.

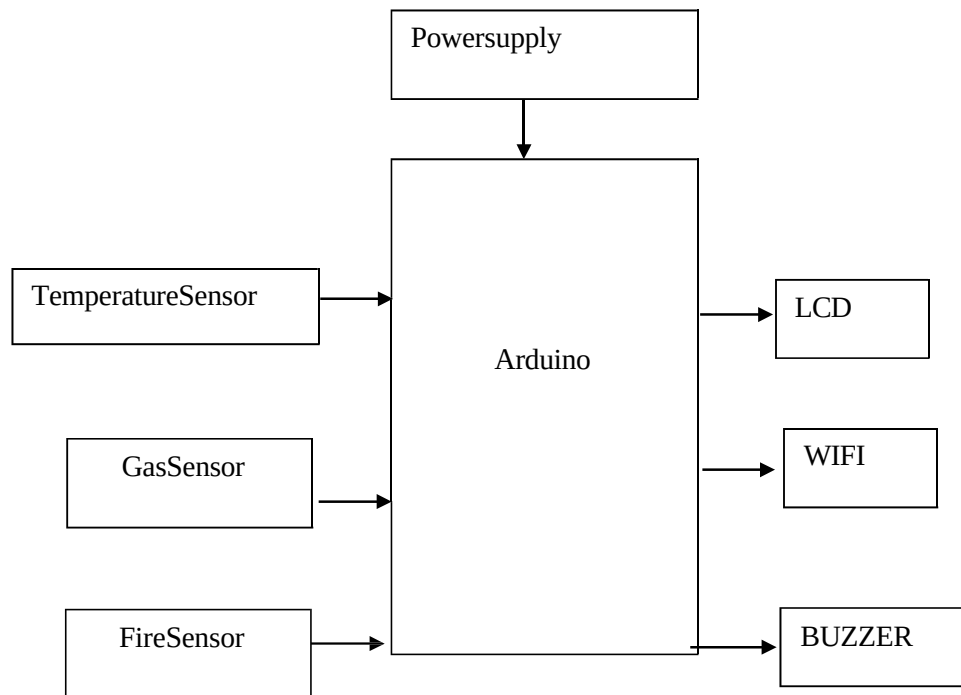
5. Environmental Interference:

- Harsh underground conditions may affect sensor performance over time (e.g., dust, moisture).

2.5 COMPARISON: EXISTING SYSTEM VS PROPOSED IOT-BASED SYSTEM

Feature	Existing System	Proposed IoT-Based System
Monitoring Method	Manual inspection and periodic checking	Real-time automatic monitoring using IoT sensors
Hazard Detection	Delayed or reactive (after an accident occurs)	Proactive detection of gas leaks, temperature rise, and other hazards
Communication	Limited or no remote communication	Wireless communication (Wi-Fi, ZigBee, or LoRa) with cloud/server
Response Time	Slow response due to human dependency	Fast response with automated alerts and notifications
Data Logging	No or minimal data logging	Continuous data logging with timestamp and cloud storage
Alert System	Manual alarm systems (e.g., sirens, lights)	Automatic alerts via SMS, mobile app, or buzzer
Accuracy & Reliability	Prone to human error	High accuracy with calibrated sensors
Safety Assurance	Low – depends on periodic manual checks	High – constant monitoring ensures early hazard detection
Cost of Operation	High – requires manpower for regular checks	Moderate – Initial setup cost but low maintenance and labor costs
Scalability	Difficult to scale	Easily scalable by adding more sensors and devices
Accessibility	Data not easily accessible remotely	Remote access through dashboard, app, or web interface
Maintenance	Requires physical inspection for faults	Can diagnose issues remotely, predictive maintenance possible

3.1 BLOCK DIAGRAM



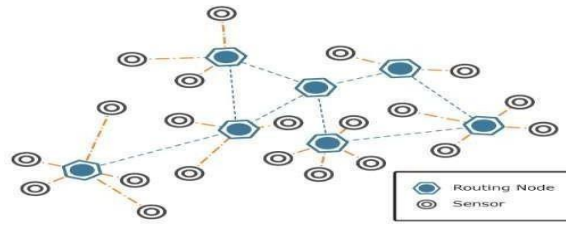
This makes it difficult for workers in the coal mining industry to obtain work. Underground coal mine explosions continue to occur despite advances in technology that allow mining monitoring methods to become more sophisticated. The harsh environments and unsafe working conditions are the primary causes of accidents and disasters in coal mines. This increases the importance of using mine checking systems in coal mines. In the proposed model, less power is consumed and efficient sensors are used to detect a worker's pulse, sweat level, and hazardous gases released into the environment. Observation systems constructed of wires have aided mine safety for many years. Wireless networks are greatly preferred in the present scenario due to the complexity and feasibility of wired networks. The main goal of this proposal is to monitor mineworker health parameters. Zig bee technology, which establishes wireless communication and updates the data from the sensor to the webpage, is used by the IoT to send the data from the mine worker section to the webpage.

Safety measures in port operations are primarily concerned with the protection working machinery, with sensors and switches used to ensure worker safety. Port worker observation and control are not possible with current technologies due to their high mobility and risk. Coal has always been India's main source of energy,

contributing considerably to the country's rapid industrial growth. It accounts for roughly 70% of total electricity production. As a result, coal's significance in the energy industry is unquestionable. However, the manufacturing process generates other byproducts, which pose a possible danger to the ecosystem and the people involved. Instead, the current effort is a genuine attempt at analyzing the gravity and developing a real-time detection tracking system based on ZigBee technology.

Wireless sensor networks (WSNs) are gaining a lot of interest around the globe right now. A WSN is a type of ad hoc, multi-hop, self-organizing network that comprises of a large number of nodes distributed over a large region to monitor the phenomena of interest. It has potential uses in medicine, the environment, science, and the defense. Wireless sensor networks are made up primarily of sensor nodes or motes that detect a phenomenon and base nodes that manage the network and gather data from distant nodes. Many variables impact sensor network design, including scalability, operation system, failure tolerance, sensor network topology, hardware limitations, transmission media, and power usage. Many variables impact sensor network design, including scalability, operation system, failure tolerance, sensor network topology, hardware limitations, transmission media, and power usage. Wired network monitoring systems have assisted the mine safety significantly, but it is not ideal for all types of mining environment. Area monitoring is a common application of WSNs.

In area monitoring, the WSN is deployed over a region where some phenomenon is to be monitored. A military example is the use of sensors to detect enemy intrusion; a civilian example is the geo-fencing of gas or oil pipelines. There are several types of sensor networks for medical applications: implanted, wearable, and environment-embedded. Implantable medical devices are those that are inserted inside the human body. Wearable devices are used on the body surface of a human or just at close proximity of the user. Environment-embedded systems employ sensors contained in the environment. There are many applications in monitoring environmental parameters,[12] examples of which are given below. They share the extra challenges of harsh environments



FigNo3.1.1 Wireless Sensor Networks

These small-sized sensors are less costly than conventional sensors and demand fewer computing and processor resources. These sensor nodes can sense, measure, and gather information from their surroundings, and then relay that information to the control center based on some local statistical decision process. It offers three significant benefits over wired monitoring network solutions. They are as follows:

1. There are no wires to place and simple installation in blind regions, lowering the expense of the surveillance system. To remove blind spots, the number of connections can be raised. It also provides basic dialogue and objective allocation
2. The dense nodes guarantee high-accuracy data capture and optimal data transfer, as well as the subsequent implementation of a real-time surveillance system for the underground environment.
3. Because sensor nodes have limited processing power and storage space, they are well suited for distant surveillance systems.

MOTIVATION FOR THE SELECTION OF PROBLEM

In underground mine, ventilation systems are critical to supply adequate oxygen, keeping up non-dangerous and non-lethal environments and an effective working mine. And also, people get over sweating. It may result unknown disease. To monitor the health condition of mine and primitive procedures of monitoring a mine's air can be followed back to the utilization of can arise and different creatures to ready diggers when the climate gets to be lethal. Incorporating health monitoring systems empowers a mine to insightfully roll out improvements in view of the far-reaching information given by the monitoring systems. Sudden changes in the system mare identified by the monitoring system, permitting quick move to be made. New and creating correspondence and following systems can be used to monitor mines more proficiently and transfer the information to the surface. The occurrence of disasters in coal mines is mainly due to the harsh environment and variability of working conditions. So, it makes the implementation of mine monitoring systems essential for the safety purpose. Wired network systems used to be a trend for traditional coal mines, which

have really played a significant role in safely production in coal mines. So, it is essential to have a wireless sensor network mine monitoring system, which can be disposed in such mines in order to have a safe production within.

Priyanka metal.,(2021) suggested a LoRaWAN-based system for monitoring man hole covers in smart city settings. For radio communication, they used the LoRaphysical layer, and for cloud access, they used the LoRaWAN protocol. LoRa is a Low Power Wide Area Network (LPWAN)-based Internet of Things protocol (IoT). It has numerous uses in the smart city environment and is rapidly being implemented. Pycom's LoPy4 Py sense nodes, inertial sensors, nano-gateway, TTN network, and application server were used. The accelerometer devices detect cover damage and theft by performing angle algorithms on a regular basis. When the manhole cover moves, the accelerometer sensor delivers a signal to the LoRa router, which wirelessly relays RF signals over long distances in the unlicensed frequency range. This signal is picked up by a neighboring LoRa nano-gateway that is linked to the TTN server. The data is transferred to the cloud and saved, and it alerts the maintenance staff for fix reasons. However, using LoRaWAN does not ensure network operators' confidence. This project's future work could include developing a Block chain-based solution for LoRaWAN network nodes that takes advantage of the benefits of both technologies. Block chain technology offers an open, decentralized, trustworthy, and tamper-proof system that includes an irrefutable method for verifying that transaction data exists.

3.2 DESCRIPTION OF REQUIREMENTS

- 3.2.1 ARDUINO
- 3.2.2 POWERSUPPLY
- 3.2.3 LCD
- 3.2.4 TEMPERATURE SENSOR
- 3.2.5 GASS ENSOR
- 3.2.6 Fire Sensor
- 3.2.7 WIFI
- 3.2.8 BUZZER

3.2.1 ARDUINO UNO:

The most common version of Arduino is the Arduino Uno. This board is what most people are talking about when they refer to an Arduino. The Uno is one of the more popular boards in the Arduino family and a great choice for beginners. There are different revisions of Arduino Uno, but the most recent revision (Rev3 or R3).

The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

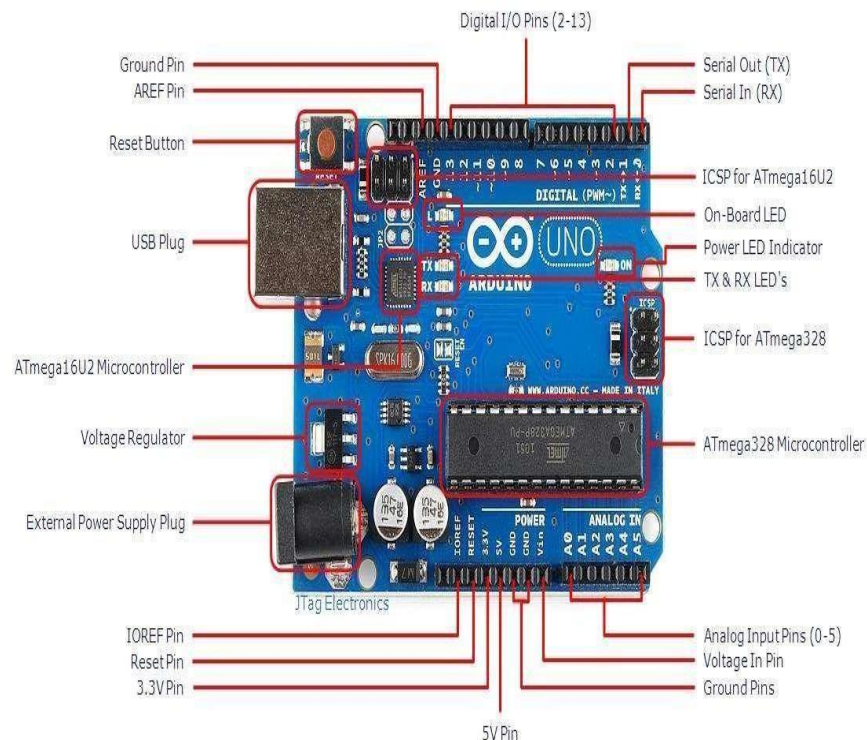


Fig No 3.2.1 ARDUINO UNO

- **Features**

Microcontroller	:	ATmega328
Operating Voltage	:	5V Input Voltage (recommended)
Digital I/O Pins	:	14(ofwhich6providePWMoutput)
Analog Input Pins	:	6
DC Current per I/O Pin	:	40mA
DCCurrentfor3.3VPin	:	50mA
Flash Memory by boot loader	:	32KB(ATmega328) of which0.5KBused
SRAM	:	2KB(ATmega328)
EEPROM	:	1KB(ATmega328)
Clock Speed	:	16MHz
Length	:	68.6mm
Width	:	53.4mm

- **USB Plug & External Power Supply Plug**

Every Arduino board needs a way to be connected to a power source. The Arduino Uno can be powered from a USB cable coming from your computer or a wall power supply that is terminated in a barrel jack. The power source is selected automatically. The USB connection is also how you will load code onto your Arduino board. The board can operate on an external supply of 6 to 20 volts.

If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

- **Voltage Regulator**

The voltage regulator is not actually something you can (or should) interact with on the Arduino. But it is potentially useful to know that it is there and what it's for. The voltage regulator does exactly what it says –it controls the amount of voltage that is let into the Arduino board. Think of it as a kind of gatekeeper; it will turn away an extra voltage that might harm the circuit. Of course, It has its limits, so don't hook up your Arduino to anything greater than 20 volts.

- **Power Pins**

Voltage In Pin – The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.

5V Pin– This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 – 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. It's not recommended. 3.3V Pin – A 3.3-volt supply generated by the on-board regulator. Maximum current draw is 50 mA.

- **Ground Pins**

There are several GND pins on the Arduino, any of which can be used to ground your circuit.

- **IO REF Pin**

This pin on the Arduino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs for working with the 5V or 3.3V.

- **Input and Output Pins**

Each of the 14 digital pins on the Uno can be used as an input or output. They operate at 5 volts. These pins can be used for both digital input (like telling if a button is pushed) and digital output (like powering an LED). Each pin can provide or receive a maximum of 40

mA and has an internal pull-up resistor (disconnected by default) of 20-5k Ohms. In addition, some pins have specialized functions.

- **Serial Out (TX) & Serial In(RX)**

Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.

- **External Interrupts**

Pins 2 and 3 can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.

PWM – You may have noticed the tilde (~) next to some of the digital pins (3, 5, 6, 9, 10, and 11). These pins act as normal digital pins, but can also be used for something called Pulse-Width Modulation (PWM). Think of these pins as being able to simulate analog output (like fading an LED in and out).

SPI – Pins 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). SPI stands for Serial Peripheral Interface. These pins support SPI communication using the SPI library.

Analog Input Pins – Labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). These pins can read the signal from an analog sensor (like a temperature sensor) and convert it into a digital value that we can read. By default they measure from ground to 5 volts, though it is possible to change the upper end of the range using the AREF pin (stands for Analog Reference. Most of the time you can leave this pin alone). Additionally, some pins have specialized functionality:

TWI – Pins A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.

- **Reset Pin**

Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

- **LED Indicators**

Power LED Indicator – Just beneath and to the right of the word “UNO” on your circuit board, there’s a tiny LED next to the word ‘ON’. This LED should light up whenever you plug your Arduino into a power source. If this light doesn’t turn on, there’s a good chance something is wrong. Time to re-check your circuit!

On-Board LED – There is a built-in LED connected to digital pin 13. When the pin is HIGH

value, the LED is on, when the pin is LOW, it's off. This useful to quickly check if the board has no problem as some boards has a pre-loaded simple blinking LED program in it. TX & RX LEDs – These LEDs will give us some nice visual indications whenever our Arduino is receiving or transmitting data(like when we're loading an reprograming to the board).

- **Reset Button**

Pushing the reset button temporarily connect the reset pin to ground and restart any code that is loaded on the Arduino. This can be very useful if your code doesn't repeat, but you want to test it multiple times.

3.2.2 POWER SUPPLY:

All digital circuits require regulated power supply. In this article we are going to learn how to get a regulated positive supply from the mains supply.

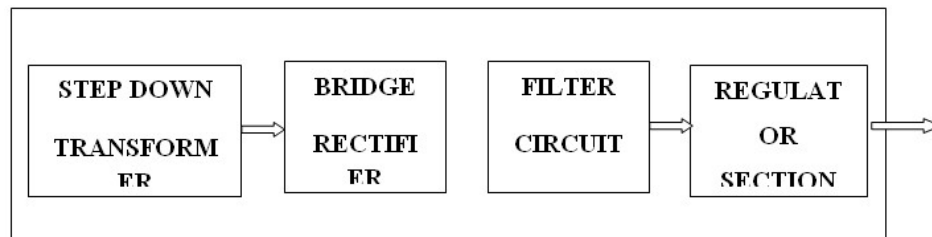


Fig No 3.2.2 Power supply

3.2.3 LCD :

LCD (Liquid Crystal Display) Shown in fig. is an electronic display module. A 16x2 LCD display is very basic module and is very commonly used in different types of devices and circuits. These modules are preferred over seven segment and other multi segment LEDs.



Fig No 3.2.3 LCD

16x2LCD

A 16x2 LCD means it display 16 characters per line and there are 2lines. By this LCD each character is displayed in 5x7 pixel matrix. This LCD contain two registers, that's are Command and Data. The command register stores the command instructions which is given to the LCD. A command is an instruction which is given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is in the form of ASCII value of the character to be displayed on the LCD.

The most commonly used Character based LCDs are based on Hitachi's HD44780 controller or other which are compatible with HD44580.

A **16x2 LCD** means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing its screen, setting the cursor position, controlling display etc. The data register stores the

data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD. LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications A16x2 LCD display is very basic module Andis very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on. A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data



Fig : LCD DISPLAY

Showing various LCD Command Description

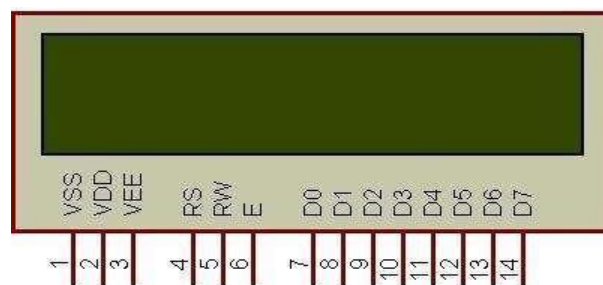


Fig: LCD Command Description

Pins of LCD

8 data pins D7:D0

Bi-directional data/command pins

Alphanumeric characters are sent in ASCII format.

- **RS: Register Select**

RS=0->Command Register is selected RS = 1 -> Data Register is selected

- **R/W: Read or Write**

0 ->Write,

1 ->Read

- **E: Enable (Latch data)**

Used to latch the data present on the data pins. A high-to-low edge is needed to latch the data.

3.2.4 BUZZER

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or electronic. Typical uses of buzzers and beepers include alarms, timers and confirmation of user input such as a mouse click or keystroke.

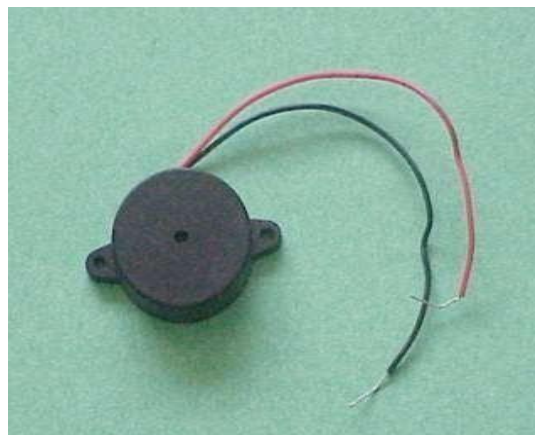


Fig No 3.2.4 **Buzzer**

2. Types Used:

For this project, a piezoelectric buzzer is commonly used. It is:

- 3 Compact and lightweight
- 4 Operates on 3V to 12V
- 5 Easy to connect with microcontrollers like Arduino or ESP32

3. Working Principle:

The buzzer converts electrical energy into sound using the piezoelectric effect. When the microcontroller detects a dangerous condition (like gas leak, fire, or high temperature), it sends a signal to the buzzer, which produces a loud alert sound.

4. Application in the Project:

- Immediate Alert: The buzzer provides an instant, loud alarm to warn miners underground.
- Offline Functionality: It works even if internet or mobile networks fail, ensuring safety.
- Multiple Triggers: It is activated by any sensor (gas, flame, temperature, vibration) that crosses a danger threshold.

5. Advantages:

- Fast and reliable local warning
- Simple to implement
- Low power consumption
- Works in real-time without internet dependency

• FEATURES

- The PB series are high performance buzzers with a unimorph piezo electric ceramic element and an integral self-excitation oscillator circuit.
- They exhibit extremely low power consumption in comparison to electromagnetic units.

Features:

- Measures Temperature Range: 0°C to 50°C
- Accuracy: $\pm 2^\circ\text{C}$
- Measures Humidity Range: 20% to 90% RH
- Operating Voltage: 3.3V – 5V
- Output: Digital signal
- Response Time: Less than 2 seconds

Working Principle:

The DHT11 sensor contains:

- A thermistor (to measure temperature)
- A capacitive humidity sensor
- An internal 8-bit microcontroller that converts the analog signal to digital
- It sends data to the microcontroller (like Arduino/ESP32) in digital format, making it easy to process and display the readings.

Application in the Project:

In a coal mine environment:

- The DHT11 continuously monitors the temperature and humidity.
- If the temperature exceeds a critical limit (e.g., 50°C), it triggers an alert to prevent heat-related hazards like fire or suffocation.
- The data is also sent to a cloud dashboard for real-time viewing by supervisors.

Advantages:

- Low cost and easy to use
- Provides both temperature and humidity in one sensor
- Low power consumption
- Good for continuous environmental monitoring

3.2.6 GAS SENSOR



Fig No 3.2.6 GAS SENSOR

Gas detectors measure and indicate the concentration of certain gases in an air via different technologies. Typically employed to prevent toxic exposure and fire, gas detectors are often battery-operated devices used for safety purposes. They are manufactured as portable or stationary(fixed) units and work by signifying high levels of gases through a series of audible or visible indicators, such as alarms, lights or a combination of signals. While many of the older, standard gas detector units were originally fabricated to detect one gas, modern multifunctional or multi-gas devices are capable of detecting several gases at once. Some detectors may be utilized as individual units to monitor small workspace areas, or units can be combined or linked together to create a protection system. As detectors measure a specified gas concentration, the sensor response serves as the reference pointer scale. When the sensors response surpasses a certain pre-set level, an alarm will activate to warn the user. There are various types of detectors available and the majority serves the same function: to monitor and warn of a dangerous gas level. However, when considering what type of detector to install, it is helpful to consider the different sensor technologies.

Working Principle:

- The MQ-135 contains a chemical-sensitive layer (SnO_2) that reacts with gases.
- When harmful gases are present, the sensor's resistance changes.
- This change is converted into a voltage signal that is read by a microcontroller (like Arduino or ESP32).
- The higher the gas concentration, the lower the sensor resistance, and the higher the output voltage.

Application in the Project:

- The MQ-135 continuously monitors the air quality in the mine.
- If gas levels cross a predefined threshold, the system triggers:
 - A buzzer alarm underground
 - A real-time alert via mobile or dashboard to supervisors
- Helps prevent gas explosions, suffocation, or poisoning by early detection

Features:

- Detectable Gases: CH_4 , CO, NH_3 , NO_x , alcohol, benzene, smoke
- Operating Voltage: 5V
- Output Type: Analog signal (some modules also provide digital output)
- Preheat Time: 20–30 seconds for stable operation

Advantages:

- Wide range of gas detection
- Fast response time
- Low cost and easy integration
- Suitable for continuous real-time monitoring

3.2.7 Flame Sensor Module

A flame sensor is a device that detects the presence of fire or flame. It is highly sensitive to light in the infrared (IR) range emitted by flames. In coal mines, detecting fire early is critical to prevent explosions and protect miners.

Sensor Type:

- 4 Most commonly, an IR-based flame sensor module is used. It typically includes:
- IR receiver tuned to flame wavelength (700–1100 nm)
 - Comparator circuit for signal processing
 - Digital and analog outputs

Working Principle:

- The sensor detects infrared light emitted by fire.
- When a flame is present within its range (usually 0–80 cm), it outputs a digital LOW signal.
- The microcontroller (Arduino, ESP32, etc.) reads this signal and triggers an alert (e.g., buzzer or notification).

Application in the Project:

- The flame sensor continuously monitors for signs of fire.
- If fire is detected, it activates the buzzer and sends a real-time alert to supervisors via mobile or cloud.
- Useful for detecting short circuits, electrical sparks, or open flames in coal mine equipment.

Advantages:

- Fast detection of visible fire
- Simple connection to any microcontroller
- Works in real-time
- Low cost and low power consumption

The pin configuration of this sensor is shown below. It includes four pins which include the following. When this module works with a micro controller unit then the pins are



Fig No 3.2.7 flame-sensor

- Pin1(VCC pin): Voltage supply ranges from 3.3V to 5.3V
- Pin2(GND): This is a ground pin
- Pin3(AOUT): This is an analog output pin (MCU.IO)
- Pin4(DOUT): This is a digital output pin(MCU.IO)

CHAPTER - 4

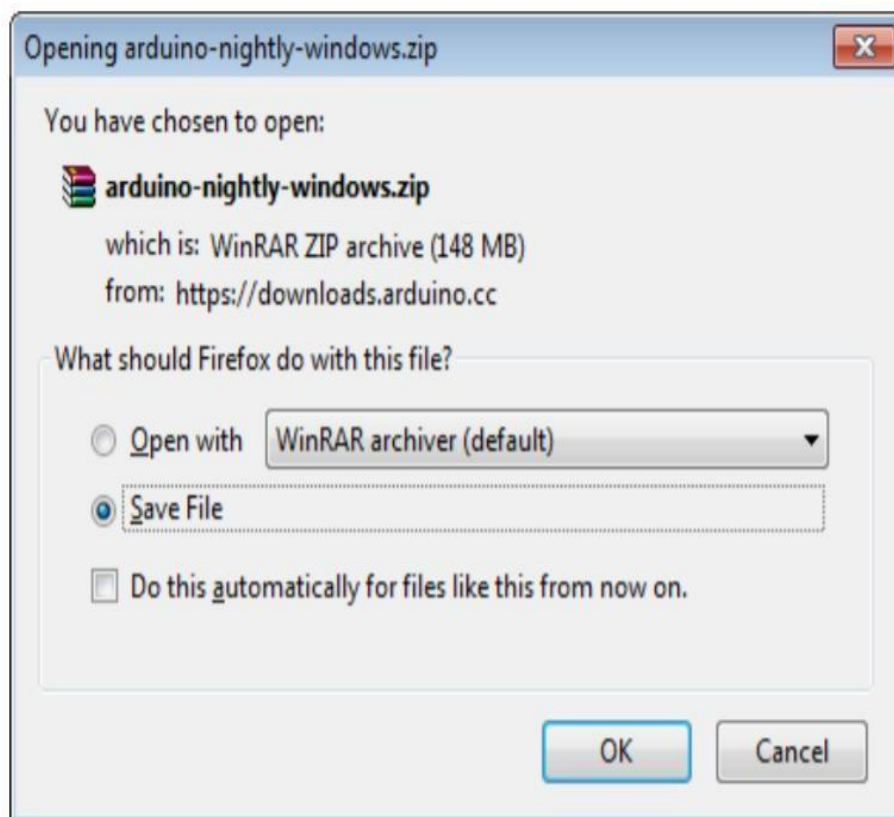
SOFTWARE REQUIREMENTS

- ARDUINO IDE
- PROGRAMMING LANGUAGE: EMBEDDED C

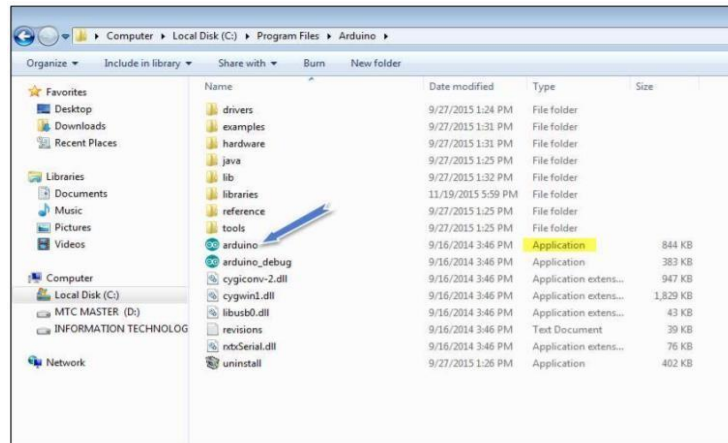
4.1 DESCRIPTION OF TOOLS

ARDUINO COMPILER (IDE)

Arduino IDE Software. You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.



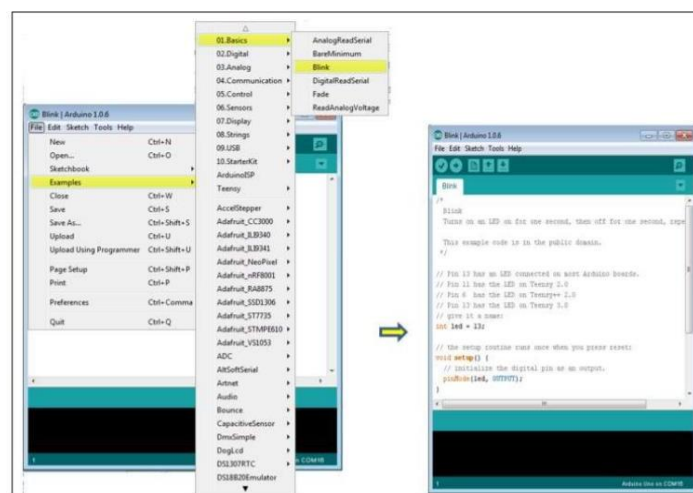
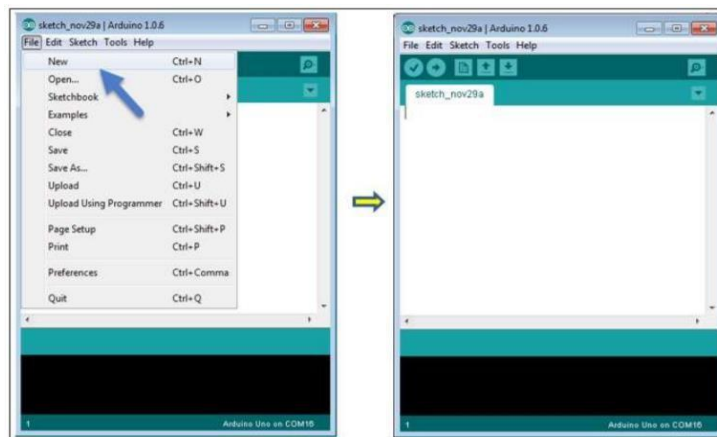
Launch Arduino IDE. After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.



Open your first project. Once the software starts, you have two options:

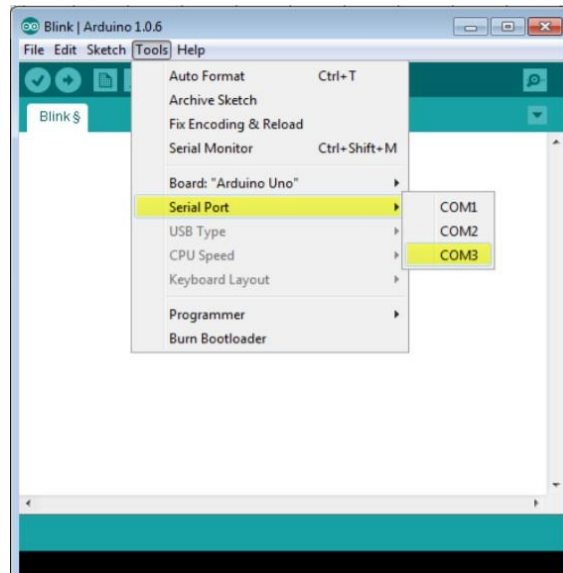
- Create a new project.
- Open an existing project example.

To create a new project, select File --> New



Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list

Select your serial port. Select the serial device of the Arduino board. Go to Tools -> Serial Port menu. This is likely to be COM3 or higher (COM1 and COM2 are usually reserved for hardware serial ports). To find out, you can disconnect your Arduino board and re-open the menu, the entry that disappears should be of the Arduino board.



Reconnect the board and select that serial port.

Before explaining how we can upload our program to the board, we must demonstrate the function of each symbol appearing in the Arduino IDE toolbar.

A-Used to check if there is any compilation error.

B-Used to upload a program to the Arduino board.

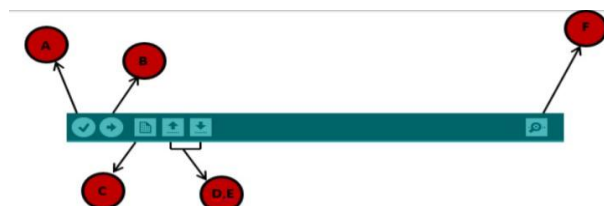
C-Shortcut used to create a new sketch.

D-Used to directly open one of the example sketch.

E-Used to save your sketch.

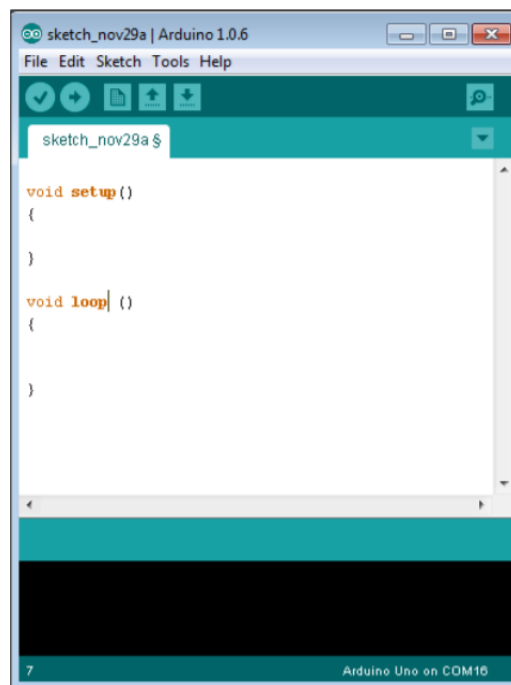
F-Serial monitor used to receive serial data from the board and send the serial data to the board.

Now, simply click the "Upload" button in the environment. Wait a few seconds; you will see the RX and TX LEDs on the board, flashing. If the upload is successful, the message "Done uploading" will appear in the status bar.



In this chapter, we will study in depth, the Arduino program structure and we will learn more new terminologies used in the Arduino world. The Arduino software is open-source. The source code for the Java environment is released under the GPL and the C/C++ microcontroller libraries are under the LGPL. Sketch: The first new terminology is the Arduino program called “sketch”. Structure Arduino programs can be divided in three main parts: Structure, Values (variables and constants), and Functions. In this tutorial, we will learn about the Arduino software program, step by step, and how we can write the program without any syntax or compilation error. Let us start with the Structure. Software structure consists of two main functions:

- Setup () function
- Loop () function



Data types in C refers to an extensive system used for declaring variables or functions of different types. The type of a variable determines how much space it occupies in the storage and how the bit pattern stored is interpreted. The following table provides all the data types that you will use during Arduino programming.

RESULTS AND DISCUSSIONS

SYSTEM IMPLEMENTATION AND TESTING

IoT based coal mine safety monitoring with hazard detection & real time alert was successfully implemented using components such as

ARDUINO UNO

GAS SENSOR(MQ-135)

TEMPERATURE SENSOR(DHT11/DHT22)

FLAME SENSOR(KY-026)

BUZZER

WIFI MODULE

POWERSUPPLY



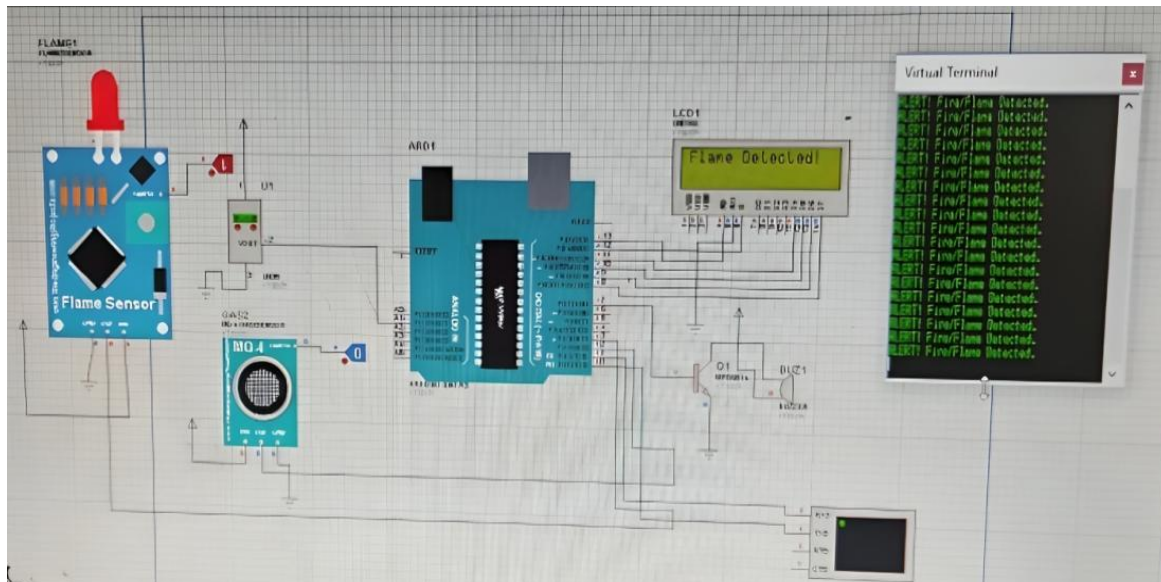
```
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code

int
gassen#include<LiquidCrystal.h>
#define buzzer 7
#define gassensor 2
#define flamesensor 3
int tempPin=1;
const int rs=13,en=12,d4=11,d5=10,d6=9,d7=8;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7);
void setup() {
    // put your setup code here, to run once:
    lcd.begin(16,2);
    pinMode(gassensor,INPUT);
    pinMode(buzzer,OUTPUT);
    pinMode(flamesensor,INPUT);
    Serial.begin(9600);
}

void loop() {
    // put your main code here, to run repeatedly:

    int tempval=analogRead(tempPin);
    float mv=(tempval/1024.0)*5000;
    float temp=mv/10;
```

The system was tested in a controlled environment by simulating different levels of impact (through abrupt movements or simulated crash scenarios). The response of the system was monitored based on sensor inputs and the accuracy of alert dispatch.

Test Scenario	Expected Outcome	Actual Outcome	Result
Gas level exceeds threshold	Buzzer sounds and alert message is sent	Alert triggered; buzzer and SMS activated	Pass
Temperature rises above safety limit	High temperature warning is triggered	Alert activated; system responded immediately	Pass
Fire detected by flame sensor	Immediate fire alert and system shutdown (optional)	Fire alert triggered correctly	Pass

KEY RESULTS & DISCUSSIONS

- **Key Results**

- 1. Multi-Hazard Detection:**

The system successfully detected harmful gases (CO, CH₄), elevated temperature levels, humidity, fire, and vibrations using respective sensors.

- 2. Fast Response Time:**

Alerts were activated within 3–6 seconds of threshold breach, ensuring timely warnings for safety.

- 3. Real-Time Alerts:**

Instant alerts were sent to the mobile app/SMS along with activation of the buzzer for on-site warning.

- 4. Offline Functionality:**

Even during loss of internet connectivity, the buzzer functioned effectively, ensuring miners were still alerted locally.

- 5. Accurate Sensor Readings:**

All sensors showed stable and repeatable values under various test conditions, confirming the system's reliability.

- 6. Live Monitoring Dashboard:**

Sensor data was displayed in real time on a cloud dashboard, accessible remotely by supervisors.

- 7. Low Power Operation:**

The complete system operated efficiently on a rechargeable battery, sustaining over 12 hours of continuous monitoring.

- 8. Compact and Modular Design:**

Hardware was compact, lightweight, and easy to mount in different sections of a coal mine.

Discussions

- The system demonstrated high reliability and responsiveness in detecting hazardous conditions in a simulated mining environment, confirming its practical usability.
- The inclusion of multiple sensors ensured comprehensive environmental monitoring, which is vital for early hazard prediction and prevention.
- The fast alert mechanism, both via app/SMS and buzzer, can play a major role in reducing risks and injuries by enabling immediate evacuation or emergency response.

- The cloud-based dashboard allowed supervisors to monitor multiple sections of the mine remotely, improving decision-making and operational control.
- The system's offline capability is a crucial advantage in deep mining environments where internet signals can be weak or absent.
- The use of low-cost components like MQ-135, DHT11, and basic microcontrollers makes this project financially viable for wide adoption, especially in small- and medium-scale mines.
- The test results confirmed that this IoT-based solution can serve as a real-time, scalable, and smart safety system to reduce coal mine accidents and fatalities.

ADVANTAGES AND APPLICATIONS

Advantages

1. Real-Time Monitoring

- Continuously tracks critical environmental parameters like gas levels, temperature, humidity, and vibrations.
- Enables early hazard detection and faster decision-making.

2. Multi-Hazard Detection

- Capable of detecting multiple risks such as methane leakage, fire, high temperature, carbon monoxide, structural shifts, and more.
- Provides a comprehensive safety solution.

3. Instant Alerts

- Sends real-time alerts through buzzers, mobile apps, or SMS, enabling immediate response to threats.
- Reduces the time gap between hazard detection and action.

4. Remote Access and Control

- Supervisors and safety personnel can monitor the system remotely through a smartphone or web dashboard.
- Enhances convenience and safety oversight.

5. Low Cost and Energy Efficient

- Utilizes affordable, low-power IoT modules (like ESP32/Arduino).
- Cost-effective for small and medium-sized mining operations.

6. Scalable and Flexible Design

- Easy to add new sensors or monitoring nodes as needed.
- Suitable for various mine sizes and configurations.

7. Data Logging and Analysis

- Stores environmental data for later review and pattern analysis.
- Supports compliance with safety audits and future planning.

- **Applications**

1. Coal Mines

- Real-time hazard monitoring for gases, temperature, and structural instability.
- Enhances underground worker safety and compliance with mining regulations.

2. Other Underground Mines

- Applicable in gold, copper, zinc, or salt mines with similar safety concerns.
- Can be customized for different environmental threats.

3. Tunnels and Underground Construction

- Monitors air quality, vibrations, and heat in subway or tunnel projects.
- Improves safety during excavation and drilling operations.

4. Oil and Gas Pipelines

- Detects gas leaks or temperature changes along pipelines.
- Sends alerts to prevent accidents and environmental damage.

5. Industrial Plants

- Monitors factory floors for gas leaks, high temperatures, or fire hazards.
- Ensures safer working environments for operators.

6. Warehouses and Storage Facilities

- Detects fire, temperature rise, or gas presence in chemical or coal storage areas.
- Prevents inventory loss and explosion risks.

7. Disaster Response Systems

- Can be deployed temporarily in landslides, building collapses, or confined rescue zones.
- Provides real-time data to rescue teams for safe entry.

CONCLUSION

The proposed IoT-based system for real-time monitoring of environmental conditions and emergency alerts represents a significant improvement over existing systems in the coal mining industry. By utilizing wireless sensor networks, fall detection mechanisms, and continuous data transmission, this system addresses key safety concerns such as gas poisoning, temperature fluctuations, and worker injuries. The integration of IoT technology enables timely interventions in case of emergencies, ensuring that miners receive the necessary help as quickly as possible. While there are still some technical challenges to overcome in terms of signal reliability and system integration, the proposed system provides a comprehensive and scalable solution for enhancing industrial safety in underground mining. With further refinement and testing, this system has the potential to drastically reduce the risks faced by miners and improve overall safety standards in the industry.

FUTURE SCOPE

The future scope of the IoT-based coal mine safety monitoring system is extensive and offers numerous opportunities for enhancement and expansion. One significant advancement could be the integration of artificial intelligence (AI) and machine learning techniques to analyze the collected data more intelligently. This would allow the system to not only detect hazards in real time but also predict potential accidents by recognizing dangerous patterns and trends before they occur.

Moreover, the deployment of wireless sensor networks (WSNs) with mesh networking capabilities can ensure wider coverage and more robust communication in the challenging underground environment, overcoming signal loss or interference issues. The system could also be enhanced by including additional sensors such as dust monitors, methane concentration detectors with higher sensitivity, and seismic activity sensors to detect underground tremors or potential collapses. Integrating automated control mechanisms, like ventilation fans or emergency shut-off valves, could help in taking immediate corrective actions without human intervention, further reducing risks. Wearable IoT devices could be introduced to monitor miners' vital signs and provide personalized safety alerts, thereby improving health monitoring alongside environmental safety.

Furthermore, advancements in energy harvesting technologies, such as vibration or thermal energy harvesters, can be employed to extend the system's battery life and reduce maintenance needs. Expanding the system for use in other hazardous industries, including chemical factories, tunnels, and oil rigs, can increase its applicability and impact. Finally, incorporating blockchain technology for secure and tamper-proof data logging could improve transparency and regulatory compliance. Overall, these future developments will make the coal mine safety system more intelligent, autonomous, reliable, and versatile, significantly enhancing worker safety and operational efficiency.

SOURCE CODE

```
int gassen#include<LiquidCrystal.h># define buzzer 7
#definegassensor2 #defineflamesensor3 int temppin=1;
constintrs=13,en=12,d4=11,d5=10,d6=9,d7=8;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7);
voidsetup(){
//putyoursetupcodehere,torunonce: lcd.begin(16,2);
pinMode(gassensor,INPUT); pinMode(buzzer,OUTPUT);
pinMode(flamesensor,INPUT); Serial.begin(9600);
}

voidloop(){
//putyourmaincodehere,torunrepeatedly:

int tempval=analogRead(temppin);
float mv=(tempval/1024.0)*5000;
float temp=mv/10; itaRead(flamesensor);
if(temp>40||gassensorval==HIGH||flamesensorval==HIGH){
digitalWrite(buzzer,HIGH);
lcd.print(" Alert");
Serial.println(" Alert!"); delay(500);
lcd.clear();
}
else{ digitalWrite(buzzer,LOW);
lcd.print("Temperature");
lcd.setCursor(0,1);
lcd.print(temp);
Serial.println("Temperature");
Serial.println(temp);
delay(500); lcd.clear();
}
}
```

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